

is continuous. Since the projective space  $\mathbb{RP}^1$  is compact, the function is bounded.

Given  $\omega = (\omega)_{n \in \mathbb{Z}} \in \Omega$ , write  $\omega = (\omega_-, \omega_+)$ , where  $\omega_- = (\omega)_{n < 0}$  and  $\omega_+ = (\omega)_{n \geq 0}$ . For one-step cocycles, the first Oseledets direction  $\mathbf{E}_1(\omega)$  depends only on  $\omega_-$ , while the second direction  $\mathbf{E}_2(\omega)$  depends only on  $\omega_+$ . So we can denote them as  $\mathbf{E}_1(\omega_-)$  and  $\mathbf{E}_2(\omega_+)$ . Let  $\mu_{\pm}$  denote the push-forward of  $\mu$  under the projection  $\omega \mapsto \omega_{\pm}$ . Then  $\mu$  coincides with the product measure  $\mu_- \otimes \mu_+$ . Therefore, by Fubini–Tonelli theorem,  $\int |\log \sin \angle(\mathbf{E}_1, \mathbf{E}_2)| d\mu$  equals the double integral

$$(32) \quad \int \int |\log \sin \angle(\mathbf{E}_1(\omega_-), \mathbf{E}_2(\omega_+))| d\mu_-(\omega_-) d\mu_+(\omega_+)$$

which is finite since the inner integral is bounded. This concludes the proof of Theorem 2 in the strongly irreducible case.

Next, we must deal with the cases where  $\Gamma_{\nu}$  permutes a finite set  $S$  of one-dimensional subspaces of  $\mathbb{R}^2$ .

If  $\#S \geq 3$ , then a cross-ratio argument shows that the projection of  $\Gamma_{\nu}$  to  $\mathrm{PGL}_2(\mathbb{R})$  is finite. This implies that  $\Gamma_{\nu}$  consists of linear conformal mappings with respect to some inner product on  $\mathbb{R}^2$ , which is incompatible with the hypothesis  $\lambda_1(\nu) > \lambda_2(\nu)$ .

If  $\#S = 2$ , then the hypothesis  $\lambda_1(\nu) > \lambda_2(\nu)$  implies that  $\{\mathbf{E}_1(\omega), \mathbf{E}_2(\omega)\} = S$   $\mu$ -a.e., and in particular the angle between Oseledets subspaces is log-integrable.

It remains to consider the case  $\#S = 1$ . Taking inverses we may assume that the stable subspace  $\mathbf{E}_2(\omega)$  belongs to  $S$  for  $\mu$ -a.e.  $\omega \in \Omega$ . With a change of basis we further assume that all  $\nu$ -almost every matrix is upper triangular. Multiplying the cocycle matrices by a log-integrable factor, we can assume that the lower right entry is always 1. That is, the function  $F$  is of the form:

$$(33) \quad F(\omega) = \begin{pmatrix} a(\omega_0) & b(\omega_0) \\ 0 & 1 \end{pmatrix},$$

where  $\log |a|$  and  $\log |b|$  are square-integrable and  $\int \log |a| d\pi < 0$ . By Lemma 5, the cocycle has Lyapunov exponents  $\lambda_1 = 0$  and  $\lambda_2 = \int \log |a| d\pi$ , and the corresponding Oseledets directions are spanned by the vectors  $(X(\omega), 1)^{\top}$  and  $(1, 0)^{\top}$ , where

$$(34) \quad X(\omega) = \sum_{n=0}^{\infty} a(\omega_{-1})a(\omega_{-2}) \cdots a(\omega_{-n})b(\omega_{-n-1}).$$

Writing  $\varphi(\omega_0) := -\log a(\omega_0)$ ,  $\psi(\omega_0) := \log b(\omega_0)$ , we have:

$$(35) \quad X(\omega) = \sum_{n=0}^{\infty} \pm e^{S_n(\omega)} \quad \text{where} \quad S_n(\omega) := \psi(\omega_{-n-1}) - \sum_{i=0}^{n-1} \varphi(\omega_{-i}).$$